CONCEPT NOTE

Integrating Machine Learning into a Bayesian Decision Network for Optimal Water Quality Sensor Placement in Urban Drainage Systems

1. BACKGROUND

This concept note outlines a proposed extension to the Bayesian Decision Network (BDN) framework for optimal water quality sensor placement in urban drainage systems developed by Sambito et al. (2020). The original framework uses the EPA Storm Water Management Model (SWMM) to simulate contamination scenarios in combined sewer networks and progressively updates a Bayesian prior to identify the optimal locations for water quality sensors. The framework was validated on two networks: the SWMM Example 8 benchmark case and the real combined sewer of Massa Lubrense, Italy, achieving isolation likelihoods of up to 84% and 92% respectively.

The primary limitation of the existing approach is computational. Convergence to a stable sensor configuration requires up to 1,000 SWMM simulation runs per sensor configuration, and the rate of convergence depends directly on the quality of the initial prior distribution. The four prior types tested in v1.0 (labelled A through D) range from non-informative uniform distributions to Monte Carlo-derived priors based on contaminant mass flux through each node, with Prior D demonstrating the fastest convergence.

The proposed work will extend this framework by replacing the Monte Carlo-derived prior with a machine learning model trained on SWMM-generated data. The ML model will learn the relationship between node-level hydraulic and topological features and the probability of detecting a contamination event, producing a data-driven prior that is expected to initialise the BDN closer to the true posterior. The central hypothesis is that this will reduce the number of SWMM simulation updates required for convergence without sacrificing sensor placement quality. The benchmark case will be SWMM Example 8, consistent with v1.0, and the same evaluation metrics (F1 and F2) will be used to enable direct comparison.

2. PROPOSED PIPELINE

Figure 1 illustrates the end-to-end workflow of the proposed Hybrid AI framework. The pipeline will proceed in nine steps, from loading the SWMM input file through to evaluation against the v1.0 baseline. Each step is described in the sections that follow.

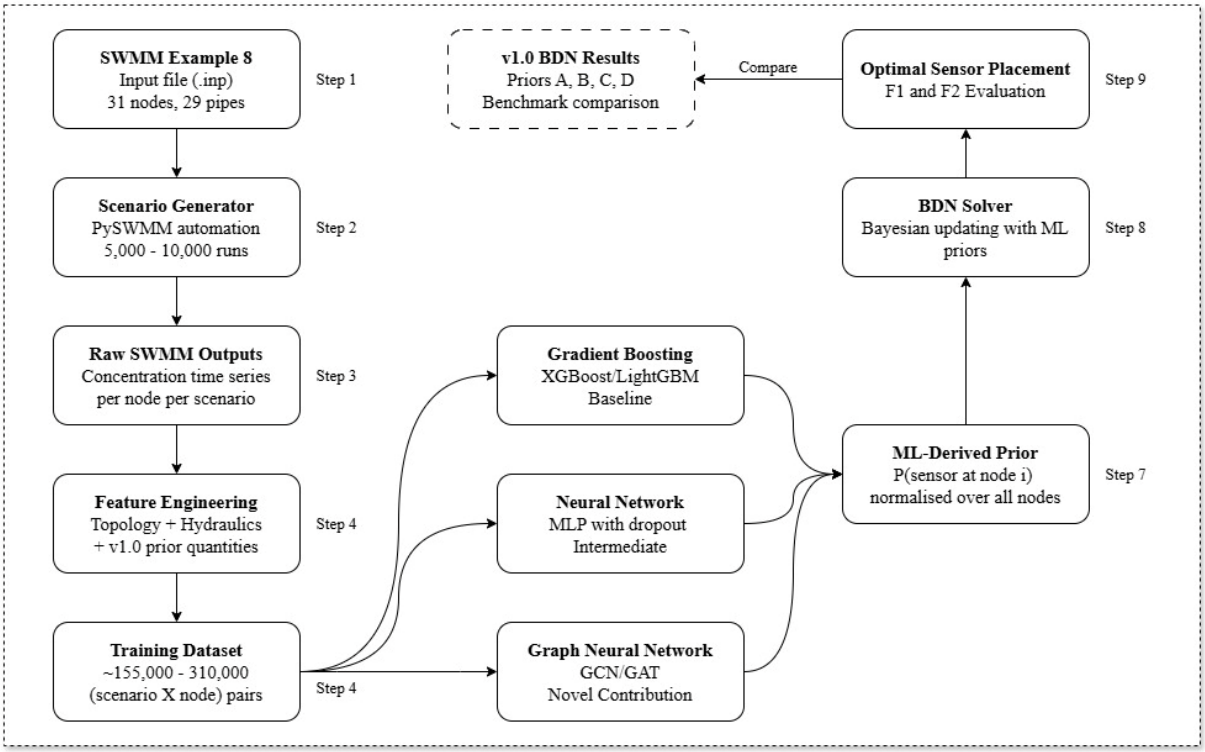

Figure 1. Proposed Hybrid AI framework pipeline.

3. TRAINING DATA -- GENERATION AND STRUCTURE

The training dataset will be generated by running SWMM programmatically across a large set of contamination scenarios using PySWMM, a Python interface to the SWMM engine. Each simulation will represent one contamination event in which the source node, contaminant mass, discharge duration, and start time are varied. The resulting pollutant concentration time series at every node in the network will be recorded and summarised into scalar features per node per scenario. Each (scenario, node) pair will constitute one row in the dataset.

Generation procedure

- Load the Example 8 input file (.inp) into PySWMM
- For each scenario: assign the injection node, contaminant mass sampled from U[0.01, 0.5] kg, duration from U[0.25, 3.0] hours, and start time; then execute the 12-hour Dynamic Wave simulation
- At each routing time step, record the pollutant concentration at every node in the network
- After each run, summarise the concentration time series at each node into scalar outputs: peak concentration, time to peak, and a binary detection flag (1 if peak exceeds 5 mg/L, else 0)
- Append one row per (scenario, node) pair to the dataset file
- Repeat across 5,000 to 10,000 scenarios, with source nodes sampled according to the non-uniform contamination probability distribution of v1.0 (nodes J4, J10, and JI18 at double the baseline probability)

Sample dataset rows

The table below shows representative rows from the proposed dataset. Each row corresponds to one network node observed during one contamination scenario. Since Example 8 has 31 nodes, each scenario

will contribute 31 rows, giving a projected total of approximately 155,000 to 310,000 rows across the full dataset.

scen_id	src_node	node_id	dist_src	topo_depth	peak_conc (mg/L)	t_peak (min)	mean_flow (m3/s)	detected
001	J4	J4	0	3	148.3	12	0.0031	1
001	J4	JI5	4	1	62.1	38	0.0084	1
001	J4	J11	7	5	4.2	71	0.0102	0
001	J4	JI10	9	0	0.0	n/a	0.0110	0
002	J10	J10	0	4	201.5	8	0.0045	1
002	J10	JI10	3	0	89.4	29	0.0110	1
002	J10	J2a	8	6	1.1	n/a	0.0028	0
003	JI18	JI18	0	2	175.0	10	0.0062	1
003	JI18	JI5	2	1	91.3	24	0.0084	1
003	JI18	J1	11	7	0.0	n/a	0.0021	0

Column notes: *dist_src* = pipe-segment distance from source node; *topo_depth* = shortest-path distance to the nearest outfall; *t_peak* = time in minutes from injection start to peak concentration (n/a where no contamination arrived); *detected* = 1 if peak concentration exceeds 5 mg/L, else 0. The node-level detection probability used as the training target will be computed per node as the proportion of all scenarios in which detected equals 1.

4. PROPOSED FEATURE ENGINEERING

The feature set will draw on three sources: the network structure available from the SWMM input file, outputs from the contamination scenario simulations, and quantities that can be derived directly from the v1.0 methodology. The inclusion of v1.0-derived quantities is principled rather than arbitrary: Sambito et al. (2020) demonstrated empirically which node-level quantities are the most informative priors, and those same quantities will be used as features in the ML model.

Group 1 -- Static topology (from SWMM input file)

- Topological depth: shortest-path distance in pipe segments from the node to the nearest outfall
- Number of upstream contributing nodes: count of all nodes from which flow can reach this node
- Betweenness centrality: frequency with which the node lies on shortest paths between all node pairs in the network graph
- Downstream path count: number of distinct routes from the node to any outfall, capturing network branching structure
- Node type: categorical variable distinguishing combined sewer junctions (Jx), interceptor junctions (JIx), the Aux3 flow-splitting node, and outfalls; v1.0 shows that JIx nodes behave systematically differently due to their position relative to flow regulators
- Flow diversion fraction: the proportion of combined flow directed to the interceptor at the nearest upstream regulator, derived from weir heights, orifice diameters, and discharge coefficients in the SWMM regulator definitions; v1.0 identifies this as the limiting factor for nodes JI10 through JI13

Group 2 -- Prior contamination probability (from v1.0, Figure 2)

The non-uniform contamination probability distribution used in v1.0 is available directly from Figure 2 of Sambito et al. (2020). Nodes J4, J10, and JI18 carry approximately double the baseline probability, reflecting known industrial activity in those sewersheds. Rather than using this distribution only as a sampling weight during scenario generation, it will also be included as an explicit feature column so that the model can learn how source probability propagates through the network to downstream nodes.

– Prior contamination probability: the node-level probability of being a pollution source, taken directly from Figure 2 of Sambito et al. (2020)

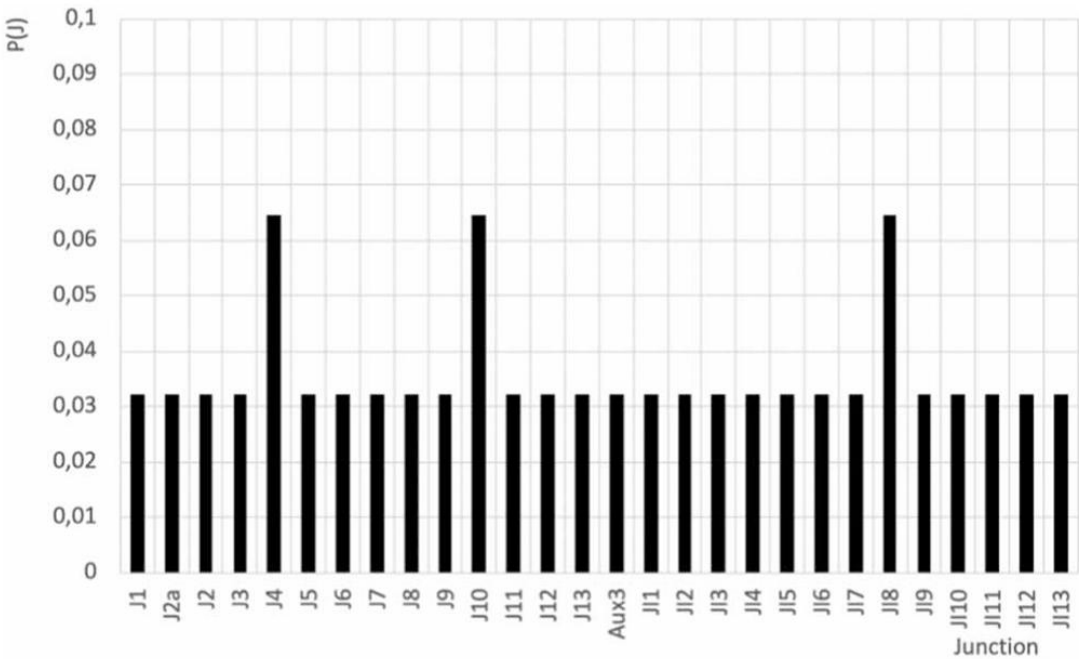

Figure 2 | Probability density function of contamination in the nodes: three nodes have a double probability of contamination.

Group 3 -- Bayesian prior quantities (from v1.0 Priors C and D)

The v1.0 framework derives two prior distributions from 50 preliminary Monte Carlo simulations: Prior C, proportional to the mean wastewater volume passing through each node, and Prior D, proportional to the mean contaminant mass through each node. Prior D is shown to be the most informative of all four priors tested, converging to the correct sensor location in 20 simulations compared to 60 for the uniform Prior A. These quantities will be computed from the same 50 preliminary runs and included as features, providing the ML model with the most discriminating hydraulic signal identified in the literature.

– Mean wastewater flux: average volume of wastewater passing through the node across 50 Monte Carlo runs; the basis of Prior C
– Mean contaminant flux: average mass of contaminant passing through the node across the same 50 runs; the basis of Prior D and the single most informative prior in v1.0
– Contaminant flux standard deviation: variance of contaminant mass across the 50 runs, capturing how consistently the node intercepts contamination regardless of source location

Group 4 -- Dynamic simulation features (from extended scenario dataset)

– Peak concentration: maximum pollutant concentration recorded at the node across all scenarios
– Mean time-to-peak: average time from injection start to peak concentration at the node, for scenarios in which contamination arrives
– Detection frequency: proportion of all scenarios in which the node exceeds the detection threshold; this is also the primary training target
– Mean flow velocity: average velocity in conduits directly connected to the node

- Concentration standard deviation: variability of peak concentration across scenarios, capturing sensitivity to source location and injection parameters

Group 5 -- Edge features (for graph-based models only)

For the Graph Neural Network architecture, node features will be supplemented by attributes on each connecting conduit, all available from the SWMM input file. These will include pipe diameter, pipe length, Manning roughness coefficient, and inlet and outlet offsets. These edge attributes will allow the message-passing layers to weight neighbour contributions according to hydraulic capacity and transport time rather than treating all connections equally.

Feature	Source	Basis in v1.0
Prior contamination probability	v1.0 Figure 2	Used as sampling prior; nodes J4, J10, JI18 at double rate
Mean contaminant flux	Prior D (50-run MC)	Most informative prior in v1.0; converges in 20 vs 60 runs
Mean wastewater flux	Prior C (50-run MC)	Second-best prior in v1.0
Contaminant flux std. deviation	Derived from Prior D runs	Captures cross-source detection consistency
Node type (J / JI / Aux / outfall)	v1.0 Figure 1	Paper shows JIx nodes behave systematically differently
Flow diversion fraction	SWMM regulator tables	Paper identifies as limiting factor for JI10 through JI13
Topological depth	SWMM network graph	Basis of Prior B pre-screening procedure
Betweenness centrality	SWMM network graph	Captures structural importance in network
Detection frequency	Extended scenario runs	Direct training target
Mean time-to-peak	Extended scenario runs	Proxy for hydraulic travel time from source
Peak concentration (mean)	Extended scenario runs	Strength of contamination signal at node

5. CANDIDATE MODEL ARCHITECTURES

Option A -- Gradient Boosting (XGBoost / LightGBM)

A gradient boosting regressor will be trained on the tabular feature matrix to predict the detection probability at each node. This architecture is well suited to structured hydraulic data, computationally efficient, and produces interpretable feature importance rankings. The importance scores will translate directly into an evidence-based explanation of why certain nodes receive higher prior probabilities, which will support the methodological narrative of the paper. This model will serve as the primary baseline for comparison.

Option B -- Feedforward Neural Network

A multi-layer perceptron with two to three hidden layers will be trained on the same tabular feature matrix. Dropout regularisation will be applied to limit overfitting given the relatively small number of distinct nodes in Example 8. This option will be pursued if the gradient boosting model shows systematic underfitting on the Massa Lubrense network, where feature interactions are expected to be more complex.

Option C -- Graph Neural Network

The sewer network will be represented as a directed graph $G = (V, E)$ where V is the set of junctions and E is the set of conduits. Node features and edge features will be propagated through message-passing layers, so that each node's learned representation encodes information from its hydraulic neighbourhood. Two

GNN variants will be evaluated: a Graph Convolutional Network (GCN), which aggregates features uniformly from neighbouring nodes, and a Graph Attention Network (GAT), which learns to weight neighbours differently based on hydraulic relevance. Implementation will use PyTorch Geometric.

A practical consideration is that Example 8 contains only 31 nodes, which is a small graph for GNN training. To address this, the GNN will initially be trained on the Massa Lubrense network (1,902 nodes) if that dataset becomes available, and performance on Example 8 will be assessed through transfer learning or fine-tuning. Alternatively, graph augmentation strategies will be explored to increase the effective training set size from Example 8 alone.

6. INTEGRATION WITH THE BAYESIAN DECISION NETWORK

The output of the ML model will be a vector of predicted detection probabilities across all N candidate sensor nodes. This vector will be normalised to sum to one and used as the prior probability distribution P(sensor at node i) in the BDN, in place of the Monte Carlo-derived Prior D of v1.0. The integration point is deliberately modular: the ML model acts as a prior generator, and the BDN solver remains unchanged from v1.0. This design allows the ML component to be substituted or improved independently, and ensures that any comparison with v1.0 priors remains methodologically clean.

The BDN will then proceed with Bayesian updating using a reduced set of SWMM-simulated contamination events, as in the original framework. The core empirical question is whether the ML prior causes the posterior distribution to stabilise at the correct optimal sensor configuration with fewer updates than Prior D, which currently requires 20 simulations for convergence on Example 8. If the ML prior is closer to the true posterior, fewer updates will be needed, reducing total computational cost.

Evaluation will compare the ML-initialised BDN against all four v1.0 prior types on three criteria: the number of SWMM simulations required to reach a stable posterior, the final sensor placement, and the detection performance metrics F1 (isolation likelihood) and F2 (detection reliability). An ablation study will further isolate the contribution of individual feature groups to convergence quality.

7. OPEN QUESTIONS

- **Massa Lubrense data.** It would be helpful to know whether the calibrated SWMM input file for Massa Lubrense is available for use in this study, as this will affect both the GNN training strategy and the scope of the validation.
- **Simulation budget.** Are there practical constraints on the number of SWMM runs, for example due to computing resources, that should guide the size of the proposed scenario dataset?
- **GNN training strategy.** Given that Example 8 has only 31 nodes, it would be worth discussing whether the GNN should be trained primarily on Massa Lubrense and transferred to Example 8, or whether a graph augmentation approach on Example 8 alone is preferred.
- **Scenario design.** Should a parallel dataset using uniform source probabilities be generated alongside the non-uniform one, to support a more controlled methodological comparison between the two experimental conditions?